

Phishing Email Investigation Report

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Case ID: PHISH-001

Executive Summary

A phishing email impersonating Banco Bradesco was analyzed to determine its legitimacy and identify potential Indicators of Compromise (IOCs). Email header analysis revealed authentication failures, including SPF, DKIM, and DMARC validation failures, indicating that the sender could not be authenticated. The sender IP address, sending domain, and email authentication records were investigated using VirusTotal, MXToolbox, and WHOIS . Based on the collected evidence, the email was classified as a phishing attempt.

Email Overview

Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje! |

Sender: BANCO DO BRADESCO LIVELO <banco.bradesco@atendimento.com.br>

Recipient: phishing@pot

Date: Tue, 19 Sep 2023 18:35:49 +0000 (UTC)

Return-Path: root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06

Message-ID: <20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>

Content-Type: text/html; charset=UTF-8

Content-Transfer-Encoding: Base64

Attachment: None identified

Email Format: HTML Email

Attack Type: Credential Phishing

Email Description

The analyzed email impersonates Banco Bradesco and attempts to create urgency by claiming that the recipient has 92,990 Livelo reward points that are about to expire. The message uses a trusted financial institution's branding to encourage the recipient to interact with the email. Header analysis identified several suspicious characteristics, including authentication failures (SPF, DKIM, and DMARC), a suspicious Return-Path, and an HTML email body encoded in Base64. These findings are consistent with phishing campaigns that attempt to bypass email security controls and deceive recipients into revealing sensitive information.

Header Analysis

The email header was analyzed to evaluate the authenticity of the sender and identify indicators commonly associated with phishing emails.

Return-Path Analysis

The Return-Path was identified as:

`root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06`

This Return-Path does not match the organization being impersonated (Banco Bradesco). Instead, it references a generic Linux server hostname, which is unusual for a legitimate financial institution. This discrepancy suggests that the email originated from an unauthorized server.

Message-ID Analysis

The Message-ID was generated by:

`<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>`

The Message-ID references the same server identified in the Return-Path. The use of a generic cloud-hosted server instead of an official Banco Bradesco mail server increases the suspicion that the message was sent from attacker-controlled infrastructure.

SPF Analysis

The email returned an SPF result of:

TempError

A DNS timeout occurred during SPF validation, preventing the recipient mail server from verifying whether the sender was authorized to send email on behalf of the claimed domain. Because the sender's identity could not be verified, the email should be treated with caution.

DKIM Analysis

The email contained the following result:

`dkim=none`

No valid DKIM signature was present. Without a DKIM signature, the integrity and authenticity of the email cannot be verified, increasing the likelihood that the message was spoofed or sent from an unauthorized source.

DMARC Analysis

The email returned the following DMARC result:

dmARC=temperror

Because DMARC validation could not be completed successfully, the receiving mail server could not confirm that the sender complied with the domain owner's email authentication policy.

Header Analysis Summary

The email header contains multiple suspicious indicators, including a mismatched Return-Path, the use of a generic server hostname, unsuccessful SPF validation, the absence of a DKIM signature, and an unsuccessful DMARC validation. While a temporary authentication error alone does not prove malicious activity, these findings, together with the impersonation of Banco Bradesco and the phishing content identified during the investigation, support the assessment that this email is a phishing attempt.

Threat Intelligence Analysis

Threat intelligence tools were used to investigate the indicators of compromise (IOCs) extracted from the phishing email. The investigation focused on the phishing URL, the sender's domain, and the sender's IP address to determine their reputation and identify any malicious indicators.

URL Analysis

The embedded phishing URL was analyzed using VirusTotal and WHOIS.

VirusTotal reported that no security vendors currently flagged the URL as malicious (0/92 detections). Although no detections were reported, the absence of detections does not guarantee that the URL is safe because newly created or low-volume phishing infrastructure may not yet be identified by security vendors.

A WHOIS lookup for the domain `blog1seguimentomydomaine2bra.me` indicated that the domain is currently not registered. This suggests that the phishing infrastructure may have been taken offline after the attack or that the domain is no longer active.

Domain Analysis

The sender domain `atendimento.com.br` was also investigated.

VirusTotal did not report any malicious detections for the domain (0/92 detections).

WHOIS records showed that the domain is publicly registered and has existed since September 20, 2018. The domain is currently in a published state and uses HostGator name servers.

Although the domain itself was not flagged by reputation services, the email still exhibited multiple phishing characteristics, including failed email authentication checks and sender impersonation. Therefore, the domain's clean reputation alone should not be considered evidence that the email is legitimate.

IP Address Analysis

The sender IP address `137.184.34.4` was previously investigated using VirusTotal, WHOIS, and MXToolbox.

The IP address belongs to DigitalOcean, LLC, a cloud hosting provider located in the United States. VirusTotal reported no malicious detections for the IP address, while MXToolbox indicated that the IP address appeared on one email blacklist. The use of cloud-hosted infrastructure combined with blacklist activity increases the suspicion that the server may have been used to distribute phishing emails.

Threat Intelligence Summary

Threat intelligence analysis did not identify widespread reputation-based detections for the phishing URL or sender domain. However, the investigation identified several suspicious indicators, including the use of cloud-hosted infrastructure, email authentication failures (SPF, DKIM, and DMARC), sender impersonation, and a blacklisted sender IP address. When correlated with the email content and header analysis, these findings support the conclusion that the message is a phishing email designed to deceive recipients into visiting a malicious website.

MITRE ATT&CK

Mapping The MITRE ATT&CK framework was used to map the observed attacker behavior identified during this phishing email investigation.

MITRE ID : T1566.002

Technique : Spearphishing Link

Evidence : Email contained a phishing hyperlink directing the victim to a fraudulent banking website.

Findings / Conclusion

The investigation determined that the analyzed email is a phishing email designed to impersonate Banco Bradesco in an attempt to steal user credentials. Multiple indicators support this conclusion. The email used social engineering by creating a sense of urgency, claiming that the recipient's Livelo reward points were about to expire. Header analysis revealed several suspicious characteristics, including a mismatched Return-Path, the absence of a DKIM

signature, unsuccessful SPF and DMARC authentication, and a Message-ID generated from a generic Linux server rather than an official Banco Bradesco mail server. Threat intelligence analysis showed that although the phishing URL and sender domain were not flagged by VirusTotal at the time of analysis, the sender IP address was associated with a cloud-hosted DigitalOcean server and appeared on an email blacklist. These findings, when correlated with the email content and authentication failures, strongly indicate a phishing campaign. Based on all collected evidence, the email was classified as a credential phishing attempt intended to deceive recipients into visiting a fraudulent website and potentially disclosing sensitive information.

Recommendations

To reduce the risk of similar phishing attacks, the following security measures are recommended:

- Do not click links or open attachments from unsolicited or unexpected emails.
- Verify the sender's identity before responding to emails requesting sensitive information.
- Implement and enforce SPF, DKIM, and DMARC to improve email authentication.
- Block or monitor suspicious IP addresses, domains, and URLs identified during the investigation.
- Use email security gateways to detect and quarantine phishing messages.
- Provide regular security awareness training so users can recognize phishing attempts and social engineering techniques.
- Report suspected phishing emails to the organization's security team for further investigation.

References

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